

# 智能阶梯轴自动上料与AI视觉检测系统项目书

60x20 mm工件 | 0.12 kg | 12件/分钟

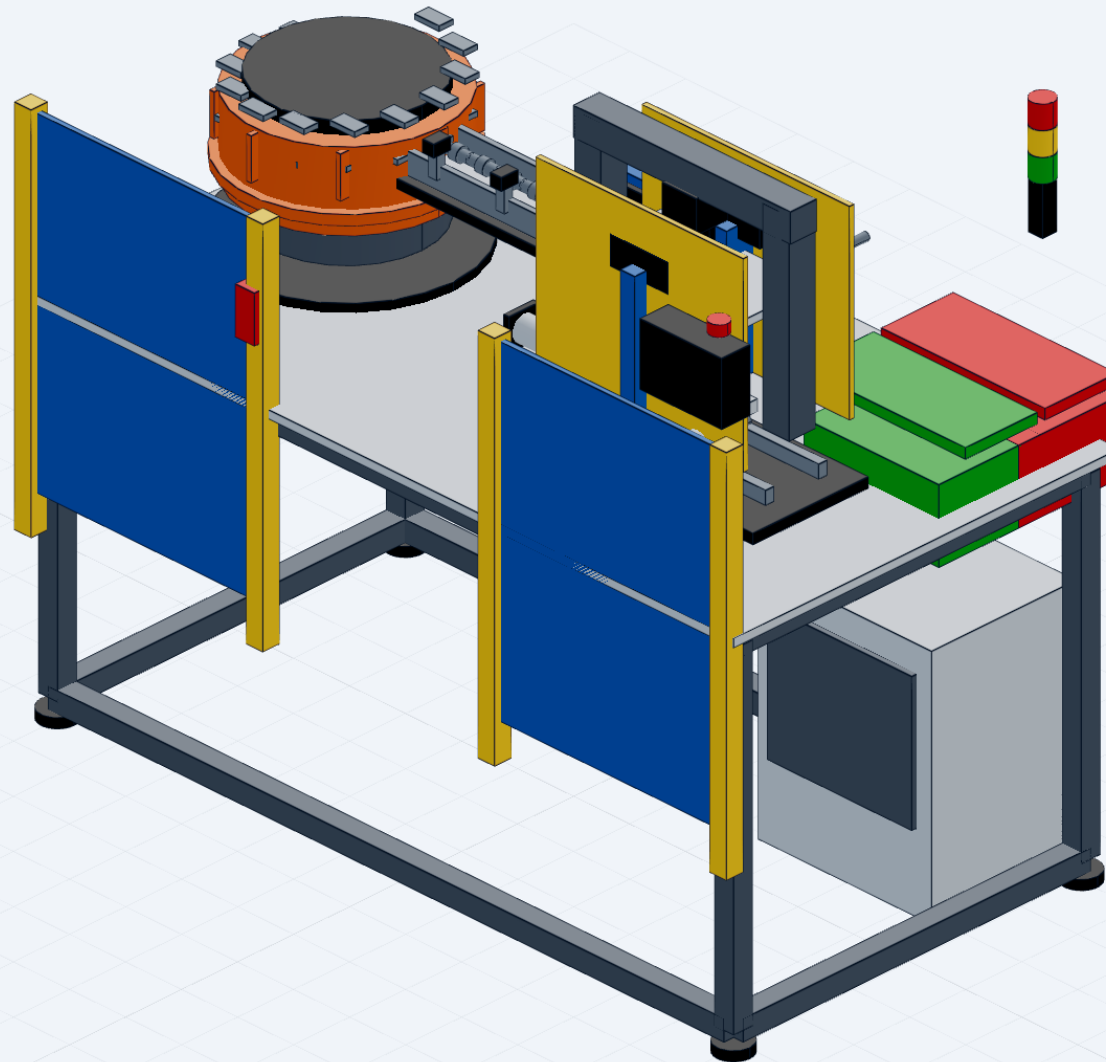
振动盘定向、直线送料、双气缸分料、V形定位、视觉/激光检测和OK/NG分流完整工作站。

|                 |
|-----------------|
| 离散螺旋轨道与可调直线送料   |
| 5 s循环动作分解       |
| 视觉轮廓与激光测径组合     |
| 阀岛、电控、安全门和防护一体化 |

Editable CAD source + STEP + GLB + calculations + BOM + manufacturing plan + validation evidence

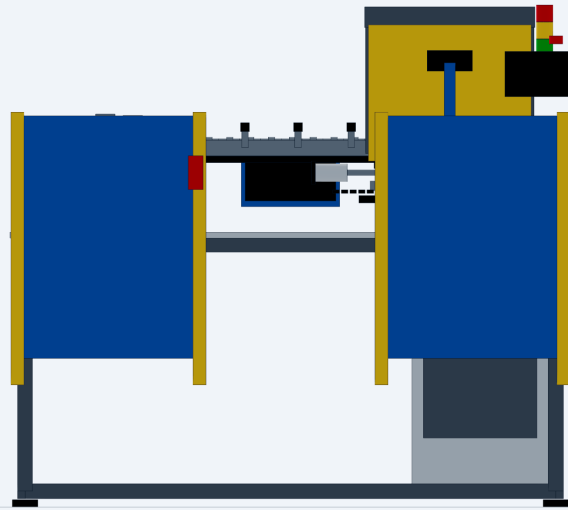
## 01 Detailed assembly overview

ISO

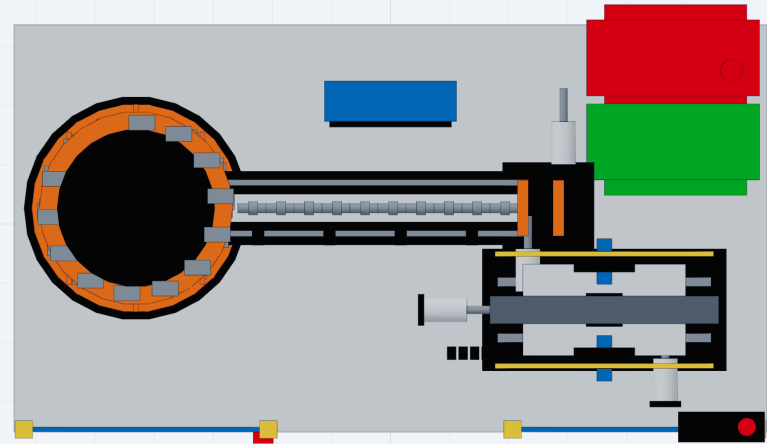


## 02 Orthographic and opposite views

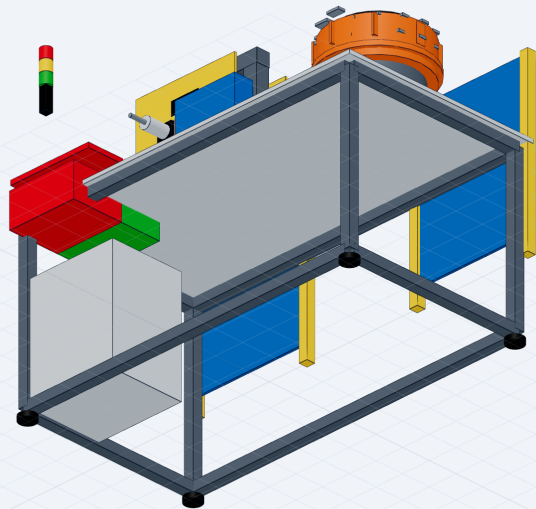
FRONT



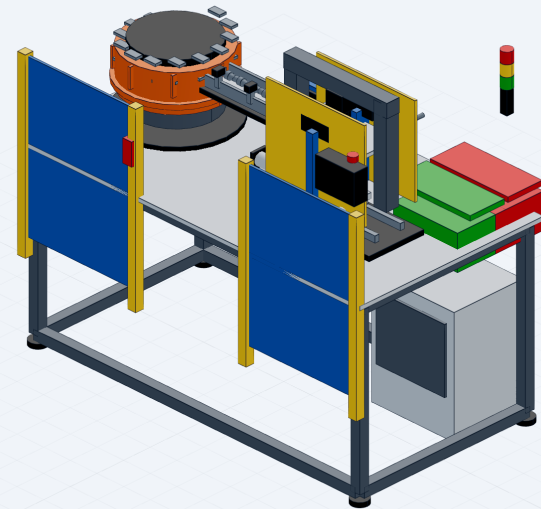
TOP



ISO



ISO



### 03 Parameters and preliminary calculations

| parameter               | value             | unit      | status            |
|-------------------------|-------------------|-----------|-------------------|
| workpiece_length        | 60                | mm        | design baseline   |
| workpiece_max_diameter  | 20                | mm        | design baseline   |
| workpiece_mass          | 0.12              | kg        | design baseline   |
| target_rate             | 12                | parts/min | target            |
| cycle_time              | 5                 | s         | calculated        |
| air_pressure            | 0.5               | MPa       | assumed           |
| transfer_cylinder_bore  | 20                | mm        | selected baseline |
| separator_cylinder_bore | 16                | mm        | selected baseline |
| model_bounding_box      | 1480x862.5x1324.5 | mm        | verified          |
| step_faces              | 924               | count     | verified          |
| step_edges              | 1698              | count     | verified          |

One 5 s cycle is allocated as: confirmation 0.4 s, separation 0.7 s, transfer 1.0 s, settle 0.4 s, image/laser acquisition 1.0 s, judgement 0.3 s, eject/return 1.2 s.

For a 0.12 kg part on a guide with assumed friction coefficient 0.2 and 0.5 m/s<sup>2</sup> acceleration, required force with factor 3 is below 1 N. A 16 mm bore cylinder at 0.5 MPa provides about 100 N theoretical force; a 20 mm transfer cylinder provides about 157 N. This leaves margin for seals, guide friction and occasional misalignment.

Five photoelectric checks cover bowl exit, track queue, first gate, second gate and V-block arrival. Laser micrometer measures the critical diameter; the top camera checks overall contour, step position, orientation and surface-defect candidates.

## 04 BOM, manufacturing and validation

| item | component                  | qty   | specification                |
|------|----------------------------|-------|------------------------------|
| 1    | vibratory bowl feeder      | 1     | 400 mm class                 |
| 2    | linear feeder track        | 1     | 620 mm adjustable rails      |
| 3    | separator cylinders        | 2     | 16 mm bore / 65 mm stroke    |
| 4    | transfer cylinder          | 1     | 20 mm bore / 170 mm stroke   |
| 5    | eject cylinder             | 1     | 16 mm bore / 85 mm stroke    |
| 6    | linear guides              | 2     | 420 mm                       |
| 7    | V-block fixture            | 1     | hardened replaceable inserts |
| 8    | photoelectric sensors      | 5     | 24 VDC                       |
| 9    | industrial camera          | 1     | top view                     |
| 10   | telecentric lens           | 1     | matched field of view        |
| 11   | backlight                  | 1     | 320 x 180 mm                 |
| 12   | laser micrometer pair      | 1     | emitter/receiver             |
| 13   | pneumatic valve island     | 1     | 8-station                    |
| 14   | PLC/HMI/electrical cabinet | 1 set | 24 VDC control               |
| 15   | OK and NG bins             | 2     | removable                    |
| 16   | aluminium-profile frame    | 1 set | 40 series                    |
| 17   | polycarbonate guarding     | 1 set | door interlock               |
| 18   | cable chain                | 1     | transfer axis                |

### Manufacturing / assembly

Track and fixture surfaces are CNC milled with replaceable wear strips. V-block inserts are tool steel, heat treated and ground. Frame members are cut aluminium profiles with dowel-located station plates. Sensor brackets use slots for calibration. Guard panels are polycarbonate in aluminium frames. Pneumatic tubing and electrical cables are separated and labelled for maintenance.

### Validation boundary

# 自动上料与视觉检测系统

## 设计验证补充说明（2026-07-24更新）

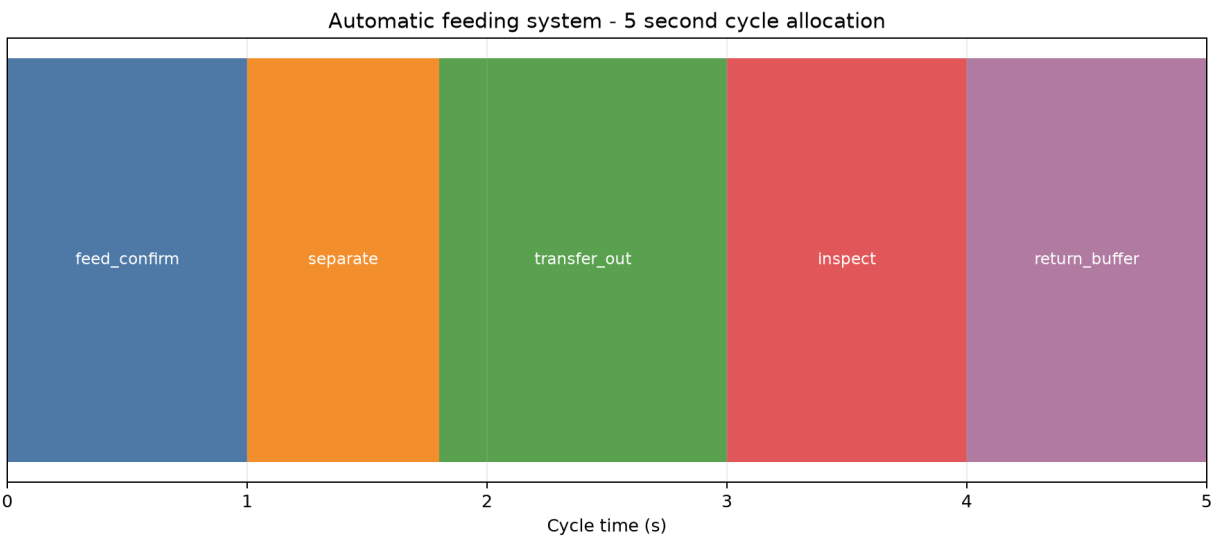
### 本次更新

针对上一版底板挠度超限完成加厚、模型更新和复算闭环。

|       |               |
|-------|---------------|
| 设计节拍  | 5 s/循环        |
| 底板优化  | 12 mm → 14 mm |
| 优化后挠度 | 0.611 mm      |
| 挠度限值  | 0.80 mm       |

### 结构复核结果

|        |                         |
|--------|-------------------------|
| 分析方法   | 一维 Euler-Bernoulli 梁有限元 |
| 材料     | 6061-T6 铝合金             |
| 等效截面   | 70 × 14 mm              |
| 最大弯曲应力 | 9.84 MPa                |
| 最大挠度   | 0.611 mm                |
| 屈服安全系数 | 24.39                   |
| 结论     | 初步通过                    |



# 实体交叠筛查与整改状态

|           |     |
|-----------|-----|
| 详细STEP实体数 | 179 |
| 包围盒候选对    | 233 |
| 精确共同体积对   | 227 |
| 求交失败数     | 0   |

## 判断说明

精确共同体积不等同于全部都是设计错误，其中可能包含轴孔配合、紧固件穿入、焊接/并体构造以及真实零件穿透。仓库已按P1/P2/P3生成整改工作表，记录空间区域推断、几何类别、处理动作和关闭状态。未完成逐零件命名和关闭前，不声明整机为无干涉制造装配。

## 原生格式状态

已验证交付包括基础SolidWorks 2023零件、增强STEP、可编辑DXF、A3 PDF、参数化重建源码、运动/节拍验证和结构复核源码。NX 12原生特征建模受本机许可证错误-10005限制；原生受约束装配和三维实体FEA尚未伪造为已完成，仓库保留失败证据与恢复步骤。

适用边界：结构结果用于方案阶段整体刚度和名义应力初筛，不能代替孔边、焊缝、轴承座、螺栓预紧、接触、屈曲、疲劳和冲击工况的三维分析。节拍、精度、视觉准确率和连续运行能力需通过最终控制系统及实体样机验证。